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BIR O'ZGARUVCHILI FUNKSIYA HOSILASI VA DIFFERENSIALI. YUQORI TARTIBLI HOSILA VA DIFFERENSIALLAR

1. t vaqtdagi ishlab chiqarish hajmi $Q = 100t - \frac{1}{30}t^3$ formula yordamida bog`langan bo`lsin. a) 5 vaqt birligiga mos; b) 10 vaqt birligiga mos mehnat unumdotligini aniqlang.

Yechish. Bu masalaning yechimini topish uchun quyidagi ishlarni amalga oshiramiz: $K'(5) = 100 - \frac{1}{10}5^2 = 97{,}5$; $K'(10) = 100 - \frac{1}{10}10^2 = 90$ ▸

2. $f(x) = \frac{x^4}{4}$ funksiya uchun [-1;1] segmentda Roll teoremasini tatbiq etish mumkinmi?

Yechish. 1. $f(x)$ funksiya uchun Roll teoremasining birinchi sharti bajariladi: $f(x)$ funksiya [-1;1] da uzluksiz;

2. $f(x)$ funksiya uchun ikkinchi shart ham bajariladi. $f'(x) = x^3$ hosila mavjud;

3. $f(x)$ funksiya uchun $f(-1) = f(1) = \frac{1}{4}$ tenglik o'rinli. Demak, Roll teoremasini tatbiq etish mumkin. $f'(c) = 0$ bo'ladigan nuqta mavjud: $f'(x) = x^3 = 0,\ x = c = 0$ da o'rinli. $f'(c) = f'(0) = 0.$

$K'(10) = 100 - \frac{1}{10}10^2 = 90.$ ▸

3. $y = x^2$ parabolaning qaysi nuqtasiga o'tkazilgan urinma $A(-1;1)$ va $B(3;9)$ nuqtalarni birlashtiruvchi vatarga parallel bo'ladi?

Yechish. $a=-1;\ b=3$. AB vatarning burchak koeffitsienti

$$k=\frac{9-1}{3+1}=2.$$

$f'(x)=2x;\ 2x=2$ tenglik faqat $x=1$ bo'lganda o'rinli, demak $x=1$ nuqtaga o'tkazilgan urinma vatarga parallel. ▸

4. Quyidagini taqribiy hisoblang: $arcsin\ 0{,}51$

Yechish. $y=arcsin\ x$ funksiya uchun $x=0{,}5$ va $\Delta x=0{,}01$ deb olamiz. Taqribiy hisoblash formulasiga ko'ra topamiz.

$$\arcsin(x+\Delta x)\approx\arcsin x+(\arcsin x)'\cdot\Delta x$$

$$\arcsin 0.51\approx\arcsin 0.5+\frac{1}{\sqrt{1-(0.5)^2}}\cdot 0.01=\frac{\pi}{6}+0.011=0.513.$$ ▸

Sonli ifodalarni 10^{-3} aniqlikda taqribiy hisoblang

5. $a\sqrt[3]{29}$.
6. $\cos 41^0$.
7. $\sqrt[7]{129}$.

8. $y=\sqrt{\ln^2 x-4}$ funksiyaning 2-tartibli hosilalarini toping.

Yechish. $y'=\dfrac{\ln x}{x\sqrt{\ln^2 x-4}}$

$$y''=\frac{4\ln x-4-\ln^3 x}{x^3(\ln^2 x-4)^{\frac{3}{2}}}$$ ▸

Funksiyalarning 2-tartibli hosilalari topilsin:

9. $y = \sin x;$

10. $y = tgx;$

11. $y = \sqrt{1+x};$

12. $y = e^{x^2};$

13. $y = ctgx.$

14.Oshkormas ko‘rinishda berilgan funksiya hosilasini toping:

Yechish. $x^3 + \ln y - x^2 e^y = 0$

$$3x^2 + \frac{y`}{y} - x^2 e^y y` - 2xe^y = 0, \quad ya'ni \quad y` = \frac{\left(2xe^y - 3x^2\right)y}{1 - x^2 ye^y}$$

Oshkormas funksiyalar berilgan. y'' toping

15. $x^2 + y^2 = a.$

16. $x + y - xy = 1.$

17. $x^3 - y^3 - 3axy = 0.$

18. $arctgy - y + xy^2 = 0.$

19. $\frac{x^2}{a} + \frac{y^2}{b} = 3.$

20.Parametrik ko‘rinishda berilgan quyidagi funksiyani differensiallang:

Yechish. $\begin{cases} y = Rsint \\ x = Rcost \end{cases}.$

$$\frac{dy}{dx} = -\frac{cost}{sint} = -ctgt$$

Tenglamalardan $\frac{d^2y}{dx^2}$ topilsin:

21. $\begin{cases} x = a\cos t, \\ y = a\sin t. \end{cases}$

22. $\begin{cases} x = t^2, \\ y = \frac{1}{3}t^3 - t. \end{cases}$

23. $\begin{cases} x = a(t - \cos t), \\ y = a(1 - \sin t). \end{cases}$

24. $\begin{cases} x = t^2, \\ y = t^3 + t. \end{cases}$

25. $y = x^2$ funksiyaning teskarisi $x = \sqrt{y}$ funksiyasi hosilasini toping

Yechish. $\frac{dy}{dx} = 2x, \quad \frac{dx}{dy} = \frac{1}{2x} = \frac{1}{2\sqrt{y}}.$

Teskari funksiyaning hosilalarini toping:

26. $y = \sqrt{x}$.

27. $y = \ln\sqrt{1 + x^3}$.

28. $y = x + \ln x$.

29. $y = x + x^3$.

30. $\lim\limits_{x\to\infty}(1 - x)^{\frac{1}{x}}$ limitni hisoblang.

Yechish. $y = (1 - x)^{\frac{1}{x}}$ deb belgilab, tenglikning ikkala qismini logarifmlaymiz

$$\ln y = \frac{1}{x}\ln|1-x| = \frac{\ln|1-x|}{x}$$

Endi limitga o'tamiz

$$\lim_{x\to\infty}\ln y = \lim_{x\to\infty}\frac{\ln|1-x|}{x} = \left\{\frac{\infty}{\infty}\right\} = \lim_{x\to\infty}\frac{\frac{-1}{|1-x|}}{1} = 0$$

$$\ln y = 0,\ y = 1 \text{ demak, } \lim_{x\to\infty}(1-x)^{\frac{1}{x}} = 1$$

Quyidagi funksiyalar limitini hisoblang:

31. $\lim\limits_{x\to 4}\dfrac{x^2-16}{x^2-6x+8}$.

32. $\lim\limits_{x\to a}\dfrac{e^x-e^a}{x-a}$.

33. $\lim\limits_{x\to 0} x\ln x$.

34. $\lim\limits_{x\to\infty}\dfrac{\ln x}{x}$.

35.Firmaning daromadi $TR=50Q-5Q^2$ funksiya ko'rinishida ifodalangan. Firmaningmarjinaldaromadini$Q=3$uchunaniqlang.

Yechish. Yuqoridagi birinchi tenglikka asosan topamiz.

$$MR = (TR)'_Q = (50Q-5Q^2)'_Q = 50-10Q.$$

$$MR(Q=3) = 50-30 = 20.$$

36.Talab funksiyasi $D=\sqrt{240-P}$ ko'rinishga ega. Talabning narxga nisbatan egiluvchanligi $P=176$ p.b uchun toping.

Yechish. $E(D)=P\cdot\frac{1}{-2(240-P)}$. $P=176$ ni qo'yamiz

$E(D)=-\frac{176}{128}=-1{,}375$. $|E(D)|=1{,}375>1$ talab egiluvchan.

QO'SHIMCHA MASALALAR

37. t vaqt oralig`ida aholi o`rtasida kasallikning tarqalishi

$$A(t)=200(t^{\frac{5}{2}}+t^2)$$

formula bilan aniqlanadi. Quyidagi vaqtlarda kasallar sonining o`zgarish tezligini aniqlang:

a) $t=1$; b) $t=4$; c) $t=9$.

38. Qandaydir mahsulotga talab $D(p)=\frac{25000}{p^2}-\frac{1}{5}$, $p-$mahsulot bahosi, Formula bilan aniqlansin. Talabning o`zgarish tezligi topilsin:

a) $p=10$; b) $p=25$.

39. Qandaydir mahsulotga talab $D(p)=\frac{100}{\sqrt{p}}-\frac{1}{4}$, $p-$mahsulot bahosi, formula bilan aniqlansin. Talabning o`zgarish tezligi topilsin:

a) $p=100$; b) $p=15$.

40. Mahsulotni sotishdan olingan foyda $R(x)=75x-0{,}05x^2$ formula bilan aniqlansin. $0\le x\le 750$, $x-$sotilgan mahsulotlar soni, foydaning o`zgarish tezligi topilsin:

a) $x=100$; b) $x=250$.

41. Quyidagi funksiyalarning o`zgarish tezligi topilsin:

a) $R(x)=2x-0{,}01x^2$; b) $R(x)=4x-0{,}005x^{\frac{3}{2}}$;

c) $R(x)=0,2x-10^{-2}x^2-10^{-4}x^{\frac{5}{2}}$; d) $R(x)=50x-2x^3(\sqrt{x}+1)$.

42.Quyidagi funksiyalarning o`zgarish tezligining qiymati topilsin:

a) $10x+p=10,\quad p=80$; b) $\sqrt{x}+3p=50,\ p=10$;

c) $x^{\frac{3}{2}}+10p=94,\ p=38,6$; d) $2p+x+0,02x^2=1000,\ \ p=494$.

43. $f(x)=x^2-6x+100,\ x\in[1;5]$ funksiya uchun Roll teoremasi shartlari x ning qanday qiymatlarida qanoatlantiradi?

.44.Roll teoremasini $f(x)=\sqrt[3]{x^2},\ x\in[-1;1]$ funksiyaga tatbiq qilish mumkinmi?

45. $f(x)=x^3-x^2-x+1$ ko'phadning hosilasi [-1;1] segmentda haqiqiy ildizga ega ekanligini ko'rsating.

46.[1;4] segmentda $f(x)=\sqrt{x}$ funksiya uchun Lagranj formulasi yozilsin va $C(x,y)$ nuqta topilsin.

47.[a, b] segmentda $f(x)=x^2$ funksiya uchun Lagranj formulasi yozilsin va $C(x,y)$ nuqta topilsin.

48. $y=2x-x^2$ egri chiziqning AB yoyi ustida shunday $M(x,y)$ nuqtani topingki, bu nuqtaga o'tkazilgan urinma AB vatarga parallel bo'lsin: $A(1;1)$ $B(3;-3)$.

49. $[-1,2]$ kesmada $\frac{4}{x}$ va $1-\sqrt[3]{x^2}$ funksiyalarga Lagranj teoremasini tatbiq qilish mumkin emasligi ko'rsatilsin.

50. $y=x^3-3x$ egri chiziqning AB yoyining qaysi nuqtasida o'tkazilgan urinma AB vatarga parallel bo'ladi?

51. $f(x)=x^3,\ \varphi(x)=x^2$ funksiyalar uchun Koshining $\frac{f(b)-f(a)}{\varphi(b)-\varphi(a)}=\frac{f'(c)}{\varphi'(c)}$ formulasi yozilsin hamda c topilsin.

52. $f(x)=a^x(a>0)$ funksiyani x ifodaga nisbatan 3-darajali ko'phad ko'rinishida tasvirlang.